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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	XMMUM
Selected Domain	DOMAIN 1
Selected SDG Track (1–17)	4, 8
Project / Solution Name	Pravista
Name (Leader)	TAN ZHI KAI
Email & Phone (Leader)	kai64327174@gmail.com , 011-6432 7174
University Affiliation(s)	XMUM, MMU
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Across emerging economies and Malaysia, the rapid integration of artificial intelligence is outpacing workforce capabilities, creating a critical skills divide. Recent empirical research (Sidhu et al., 2024; Babashahi et al., 2024) reveals that nearly half the workforce (47.3%–49%) remains unfamiliar with AI technologies, while 35% report an acute shortage of skilled AI talent.

This crisis is driven by three systemic bottlenecks:

1. **Curricular Latency & Academic Disconnect**: Traditional higher education curricula lag behind rapid technological advancements, facing a scarcity of AI-specialized faculty and failing to prepare graduates for industry demands in data analytics, IoT, and edge automation (Babashahi et al., 2024).
2. **Critical Competency Gaps**: The most severe workforce deficits lie in Ethical and Responsible AI Use (26.0%) and AI System Integration (25.0%)—areas rarely taught in conventional degree programs.
3. **Prohibitive Training Barriers**: When individuals attempt self-directed upskilling, they encounter severe real-world constraints: lack of time (29.0%), prohibitive training costs (26.0%), and inaccessible quality curricula (24.0%) (Sidhu et al., 2024).

Furthermore, traditional recruitment systems evaluate applicants through opaque ATS algorithms without actionable diagnostic feedback. Without an accessible, explainable bridge between current capabilities, ethical AI literacy, and structured reskilling pathways, emerging-market talent remains vulnerable to technological displacement (SDG 8).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Pravista is an explainable career-intelligence and reskilling platform built for Malaysian students and job switchers, which includes questionnaire, CV upload and review, CV evaluation, career matching, scenario exploration, learning pathways and SDG connections. Rather than relying on black-box

generative models that hallucinate qualifications, Pravista bridges workforce intelligence through transparent, deterministic rules paired with targeted GenAI:

1. **Privacy-First CV Diagnostics**: An ephemeral in-memory document parsing engine that supports parsing of PDF, DOCX, and Tesseract OCR. It extracts capabilities while immediately discarding raw documents from memory to protect user data.
2. **Explainable Multi-Factor Matching**: The platform evaluates users against curated occupational roles mapped to World Economic Forum (WEF) 2025–2030 skills benchmarks and O*NET standards. Match scores explicitly break down technical coverage (45%), transferable skills (20%), interests (15%), work style (10%), and practical constraints (10%), ensuring zero unexplained rejections.
3. **Integrated Classroom & Digital Badges**: A built-in hosted learning studio allows modular course study with streaming PDF or video lessons and four question types, which are single, multiple, fill-in-the-blank, and mix-match. Passing chapter quizzes unlocks final assessments that award cryptographically verified digital badges on user profiles, directly updating career-matching evidence.
4. **Purpose-Driven SDG Mapping**: Using Google Gemini, student project logs are semantically assessed against UN SDG targets (e.g., Targets 4.4, 8.2, 9.5, 13.3) to award milestone XP and generate verified CV additions with explicit user consent.

The platform integrates GenAI APIs, database architecture, as well as cloud infrastructure. The intelligence layer leverages Google Gemini with temperature-constrained, structured JSON schemas and deterministic keyword fallbacks for offline resilience. This is supported by a Supabase PostgreSQL backend of 17 relational tables secured by Row-Level Security (RLS) for tenant data isolation. Complementing this, private cloud storage delivers lesson media via short-lived signed URLs, powering digital credential badges that directly update career-matching evidence.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16 (App Router, React 19, TypeScript), Tailwind CSS, GSAP ScrollTrigger & Flip animations, TanStack React Query v5, Recharts, @xyflow/react (interactive SDG knowledge graph), and pdfjs-dist (sandboxed client document preview).

Backend: Python 3.12, FastAPI, Uvicorn, Pydantic v2 data contracts, PyMuPDF, python-docx, PyJWT, and Tesseract OCR.

Database & Cloud Storage: Supabase PostgreSQL with 17 relational tables, native Row-Level Security (RLS) policies, session pooler (port 5432), and private Supabase Storage buckets utilizing 6-hour signed bearer URLs.

Architecture Flow:

Client (Next.js) ➔ REST/JWT ➔ Backend (FastAPI) ➔ Ephemeral Parsing & Deterministic Rubric Scoring ➔ Supabase (PostgreSQL with RLS / Private Storage) ➔ Optional Bounded Google Gemini / OpenAI JSON Schema calls.

Yes. The platform utilizes the Google Gemini API (generateContent) for semantic project SDG assessment and role summaries, deployed alongside Supabase cloud database and private storage infrastructure.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied user-centered design thinking to address student anxiety around career readiness

Empathize & Define: Research showed graduates distrust opaque ATS algorithms. We designed an explainable UI that exposes exact mathematical rubric weights, benchmark citations (WEF, UN DESA), and actionable skill gaps.

Ideate & Prototype: We created zero-login fictional demo personas (Aisha for undergraduates, Daniel for career switchers) enabling instant judge evaluation without registration hurdles. We built an accessible, responsive UI with light/dark theme persistence and WCAG-compliant contrast.

Test & Validate: The architecture is verified by 57 frontend component tests, 35+ backend API integration tests, and Playwright end-to-end journeys covering cross-user isolation, offline network recovery, and token lifecycles.

Deployment Strategy: The system is built on a decoupled Next.js and FastAPI containerized architecture, and is deployment-ready for cloud platforms (Vercel edge network and AWS/GCP container runtimes). All database transactions enforce strict rollback verifications, prepared-statement hardening, and PostgreSQL Row-Level Security, ensuring high availability, zero runtime crashes, and enterprise-grade data isolation for Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG Target 8.6**: "Substantially reduce the proportion of youth not in employment, education or training." **SDG Target 4.4**: "Substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship."
Current Problem Baseline	Empirical research in emerging markets (Sidhu et al., 2024) establishes that 49% of the workforce is unfamiliar

Metric Category	Your Response
What is the starting condition of the problem, expressed as a number? *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	with AI, while 35% cite a severe scarcity of skilled AI talent. The primary skill deficiencies are Ethical AI Use (26%) and AI Integration (25%), with adoption hindered by time constraints (29%), high training costs (26%), and lack of access to quality curricula (24%).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• Skill-Gap Resolution Efficiency: Time required to generate an explainable, personalized reskilling roadmap (benchmarked from weeks to < 10 minutes). • Cost-Free Access: 100% free curated learning resources bypassing the 26% cost barrier. • Verified Learning Milestones: Number of completed interactive chapter quizzes, course badges, and verified SDG project logs.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve 50% reduction in user-reported career planning time; 100% explainability on skill gap diagnostics (zero hallucinated requirements); minimum 80% passing rate on modular ethical AI and technical assessments across 500+ pilot student users.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

✅ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

✅ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

✅ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

✅ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js / React	Code	MIT License
FastAPI / Pydantic	Code	MIT License
SQLAlchemy / asyncpg	Code	MIT License
PyMuPDF (Fitz)	Code	AGPL-3.0
Google Gemini	Model	Google Terms of Service
Tesseract OCR	Model	Apache 2.0 License
WEF Future of Jobs 2025	Dataset	Open Citation Reference
O*NET Database	Dataset	CC BY 4.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[25/10/2026]